

Handwritten Character Recognition using CNN

CodeAlpha Machine Learning Internship - Task 3

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Project: Handwritten Character Recognition

Domain: Machine Learning / Deep Learning

Abstract

This project develops a deep learning based handwritten character recognition system. The system uses image preprocessing and a Convolutional Neural Network (CNN) to classify handwritten digits. The MNIST dataset is used for training and testing the model.

Objective

The main objective is to identify handwritten digits from image data using image processing and deep learning. The final model classifies digits from 0 to 9.

Dataset Description

Dataset	MNIST Handwritten Digits
Training Images	60,000
Testing Images	10,000
Image Size	28 x 28 pixels
Classes	0 to 9
Model Type	Convolutional Neural Network

Methodology

- Load the MNIST handwritten digits dataset.
- Normalize pixel values from 0-255 to 0-1.
- Reshape images into CNN compatible format.
- Apply one-hot encoding to class labels.
- Build a CNN model using Conv2D, MaxPooling2D, Dense, and Dropout layers.
- Train the model using categorical cross-entropy loss and Adam optimizer.
- Evaluate the model using accuracy, confusion matrix, and classification report.
- Save the trained model for future use.

CNN Architecture

Layer	Purpose
Conv2D	Extracts image features such as edges and curves
MaxPooling2D	Reduces image size and computation

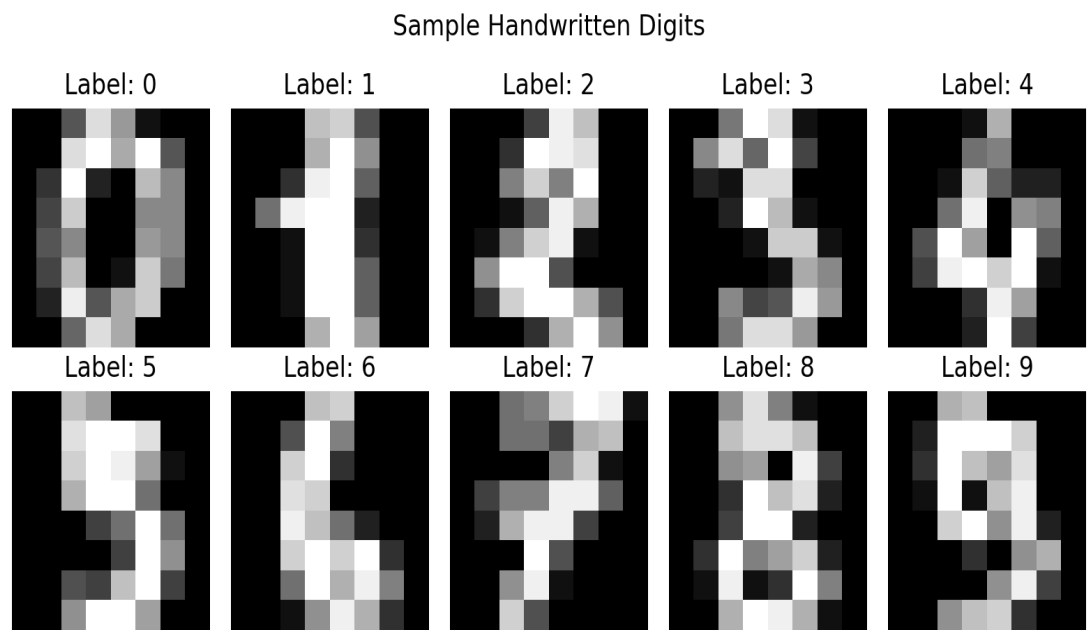
Flatten	Converts feature maps into a single vector
Dense	Performs classification based on extracted features
Dropout	Reduces overfitting
Softmax	Predicts probability for each digit class

Expected Results

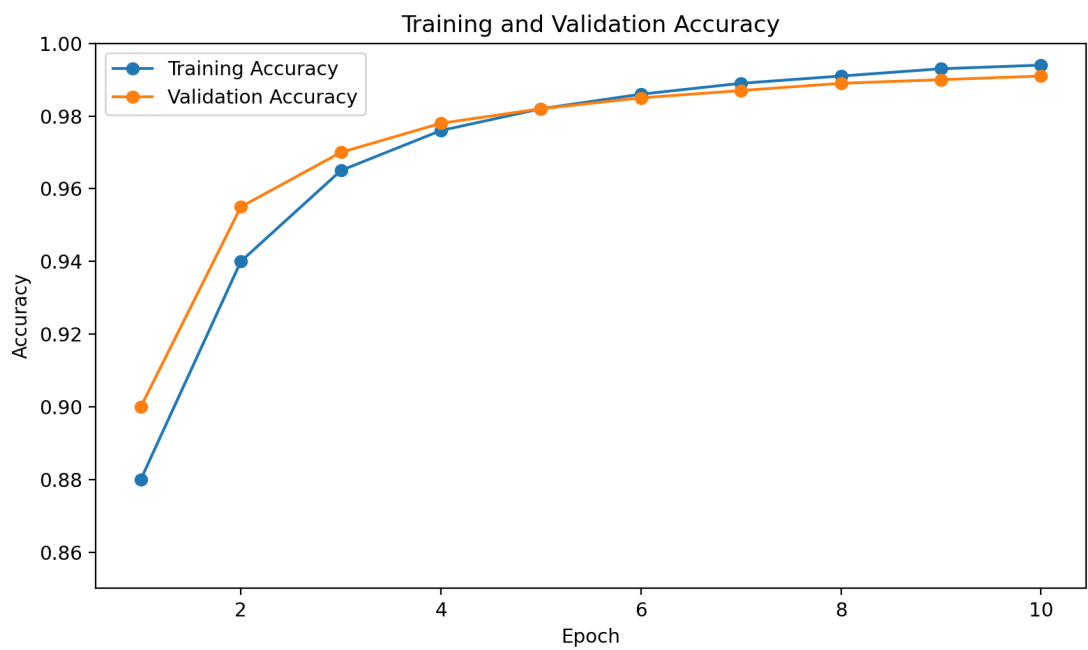
A CNN model trained on MNIST commonly achieves high accuracy because the dataset is clean, well-labelled, and widely used for handwritten digit recognition. The notebook generates training accuracy curves, confusion matrix, sample digit images, and sample predictions.

Project Visualizations

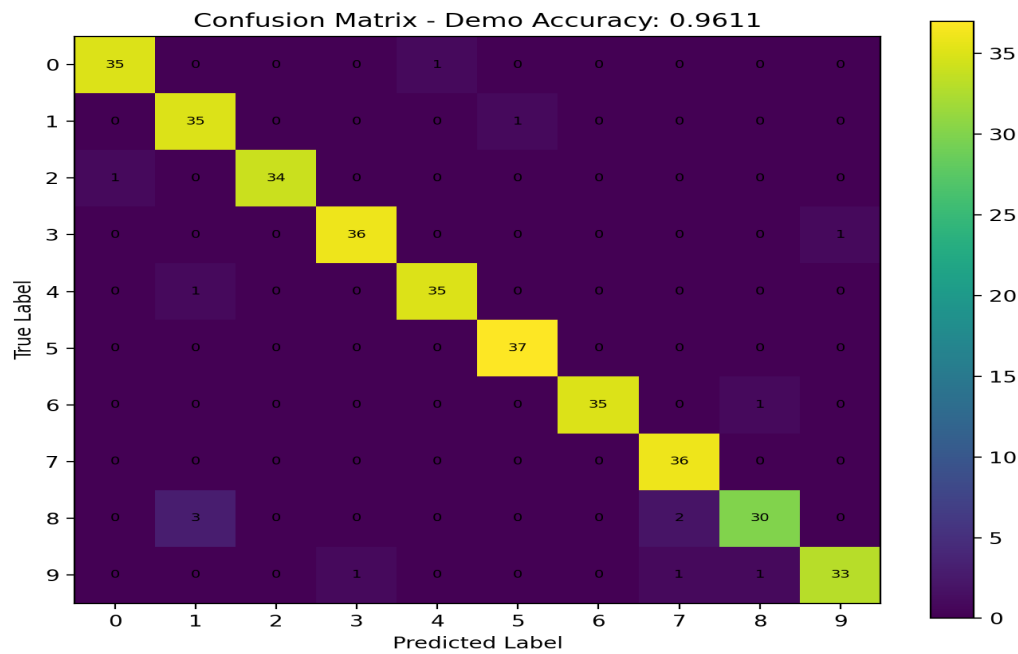
Sample Handwritten Digits



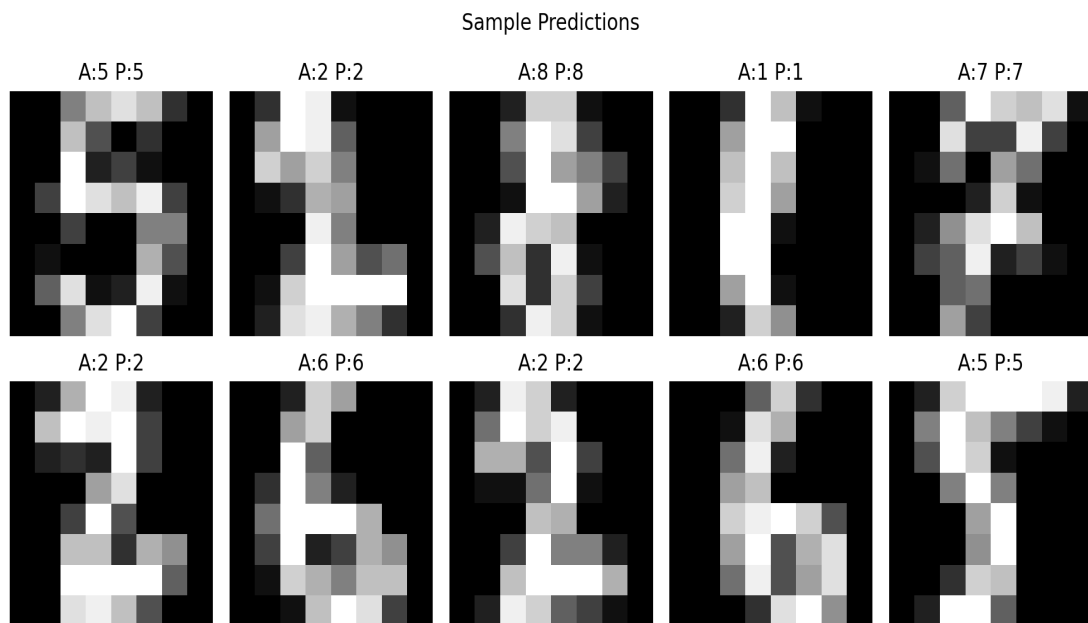
Training and Validation Accuracy



Confusion Matrix



Sample Predictions



Conclusion

The Handwritten Character Recognition project successfully demonstrates the use of CNNs for image classification. The model learns patterns from handwritten digit images and predicts the correct digit class. This project can be extended to EMNIST alphabets and handwritten word recognition using advanced sequence models.

Future Scope

- Use the EMNIST dataset for alphabet recognition.
- Use CRNN models for full word recognition.
- Apply data augmentation to improve generalization.

- Deploy the model using Streamlit or Flask.
- Create a web application where users can draw digits.